CoDrone Kit

1 kit · 18 drones

WHAT'S IN THE KIT

- 18 CoDrones
- Coded landing pads
- Extra batteries and chargers
- Spare propellers and body frames

QUICK-START GUIDE

1. Charge all batteries before class (allow ~40 min).
2. Students code flight paths in Blockly or Python via the CoDrone app/software.
3. Start with takeoff-hover-land before any movement code.
4. Fly in a gym or large clear space — set a strict launch-zone routine.
5. Rotate batteries; each flight is ~8 minutes.

FULL LESSONS IN THIS GUIDE

PRO TIP: *Establish 'props stop before hands touch' as rule one. Extra parts are included for the inevitable crash.*

Questions, co-planning, or co-teaching? Nick Schuster · nschuster@lps.k12.co.us · 303-347-4388

Request this item: LPS Library of Things website

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LESSON · GRADES 6-8

Rescue Mission Coding

LEARNING OUTCOMES

- Students will program a drone flight sequence with takeoff, waypoints, and landing
- Students will measure and iterate to improve landing accuracy
- Students will follow safety protocols for shared airspace

MATERIALS

- CoDrones with charged batteries (plus spares on chargers)
- Devices with CoDrone Blockly software
- Taped grid map with labeled target zones
- Measuring tape for scoring landing distance

INSTRUCTIONS

1. Safety brief first: launch zone lines, props-stop rule, one team flies at a time per zone.
2. Teams program: take off → hover 3 s → fly the rescue route → land on target.
3. Trial 1 flights; measure landing distance from target center.
4. Iterate: adjust durations/distances in code; log every change.
5. Trial 2 and 3; track improvement between runs.
6. Debrief: which single code change bought the most accuracy?

ACTIVITY

Rescue Mission Coding — drones fly a search-and-rescue route over a taped grid and land on target zones. Scoring by centimeters-from-target turns iteration into a competitive instinct.

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LESSON · GRADES 9-12

Autonomous Flight Trials

LEARNING OUTCOMES

- Students will write Python flight programs with parameterized movement
- Students will collect trial data and refine parameters systematically
- Students will document an engineering iteration cycle

MATERIALS

- CoDrones + batteries
- Devices with CoDrone Python environment
- Obstacle course materials (hoops, cones, tape)
- Trial log template (parameters + outcome per run)

INSTRUCTIONS

1. Course preview: teams walk the obstacle course and plan a flight path on paper.
2. Write the autonomous program in Python — no manual control allowed.
3. Every flight gets a log entry: parameters used, what happened, what changes next.
4. Enforce one-variable-at-a-time changes between trials.
5. Timed autonomous runs; score = completion + time + penalty for touches.
6. Final deliverable: the log plus a paragraph on their tuning strategy.

ACTIVITY

> Autonomous Flight Trials — a timed obstacle course flown entirely by code. The trial log requirement turns 'try stuff until it works' into disciplined parameter tuning.